3.5.1	**Thermodynamics**	
	Enthalpy change (ΔH)	be able to define and apply the terms enthalpy of formation, ionisation enthalpy, enthalpy of atomisation of an element and of a compound, bond dissociation enthalpy, electron affinity, lattice enthalpy (defined as either lattice dissociation or lattice formation), enthalpy of hydration and enthalpy of solution
		be able to construct Born–Haber cycles to calculate lattice enthalpies from experimental data. Be able to compare lattice enthalpies from Born–Haber cycles with those from calculations based on a perfect ionic model to provide evidence for covalent character in ionic compounds
		be able to calculate enthalpies of solution for ionic compounds from lattice enthalpies and enthalpies of hydration
		be able to use mean bond enthalpies to calculate an approximate value of ΔH for other reactions
		be able to explain why values from mean bond enthalpy calculations differ from those determined from enthalpy cycles
	Free-energy change (ΔG) and entropy change (ΔS)	understand that ΔH, whilst important, is not sufficient to explain spontaneous change (e.g. spontaneous endothermic reactions)
		understand that the concept of increasing disorder (entropy change ΔS) accounts for the above deficiency, illustrated by physical change (e.g. melting, evaporation) and chemical change (e.g. dissolution, evolution of CO_2 from hydrogencarbonates with acid)
		be able to calculate entropy changes from absolute entropy values
		understand that the balance between entropy and enthalpy determines the feasibility of a reaction; know that this is given by the relationship $\Delta G = \Delta H - T\Delta S$ (derivation not required). be able to use this equation to determine how ΔG varies with temperature be able to use this relationship to determine the temperature at which a reaction is feasible